

Osteoarthritis (OA) is a disorder marked by the deterioration of articular (hyaline) cartilage, the connective tissue lining the ends of bones at most movable joints. The progressive erosion of articular cartilage in OA is thought to be caused by an inflammatory response induced by years of biomechanical stress due to excessive compression and friction. This response involves the release of proteolytic enzymes that break down cartilaginous proteins. The progressive breakdown of articular cartilage ultimately results in complete exposure of the cortical portion of the underlying subchondral bone with narrowing of the joint space. Repeated and abnormal joint stress can also cause the development of osteophytes (or small, bony growths) that form on the subchondral bone. These degenerative processes cause pain and restrict joint movement.

Stem cell therapy, in which injected or grafted stem cells are induced to differentiate into chondrocytes (cartilage cells), is a potential new treatment for OA. To further understand chondrogenesis, researchers have been studying the molecules that regulate the expression and activity of the transcription factor Sox9, which has been shown to promote chondrocyte differentiation. For example, researchers have found that binding of the growth factor TGF- β to its receptor results in Sox9 activation.

MicroRNAs (miRNAs), which are known to alter gene expression by acting at the translational level, are also potential regulators of Sox9. In particular, miRNA-145 is thought to regulate Sox9 expression by binding to the 3'-untranslated region (UTR) of the Sox9 mRNA transcript. After chondrocyte differentiation was induced using TGF- β , researchers used western blotting to generate an expression profile of Sox9 and β -actin (a protein not regulated by miRNA-145) in mouse stem cells transfected with miRNA-145, anti-miRNA-145, or neither (Figure 1).

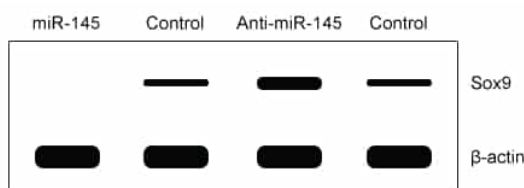


Figure 1 Expression of Sox9 in the presence and absence of miRNA-145 or anti-miRNA-145 following induction of chondrocyte differentiation

OA commonly affects large, weight-bearing joints of the lower extremities such as the knee and hip. Based on the passage, which of the following would most likely occur in a patient with OA involving the knees?

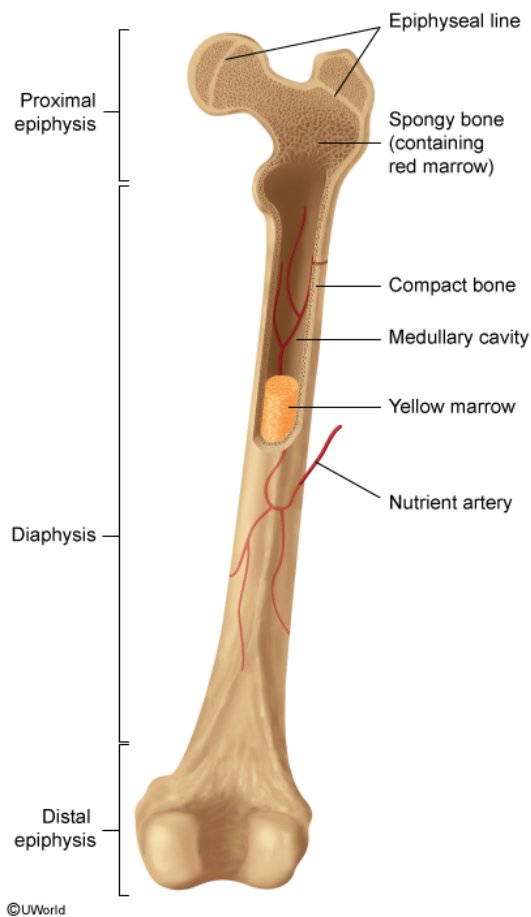
- ☐ A. Deterioration of the yellow bone marrow in affected bones [8%]
- ☒ B. Deposition of excess bone on the epiphyses of affected long bones [52%]
- ☐ C. Loss of hematopoietic function in the red marrow of affected bones [2%]
- ☐ D. Degeneration of compact bone at the diaphysis of affected long bones [36%]

Correct

52% answered correctly

Explanation:

Bone structure: compact & spongy bone



The human skeleton can be divided into axial (ie, skull, spine, ribcage) and appendicular (ie, limb) components. Long bones, one of the four bone types, are aptly named based on their appearance: their length is greater than their width. The long bones (eg, femur) serve to provide a framework for movement. The structure of long bones includes the following:

- **Epiphyses** are rounded ends that have joint surfaces covered by articular cartilage.
- The **diaphysis** is the hollow shaft (*medullary cavity*) filled with bone marrow (primarily of the yellow type in adulthood).
- **Metaphyses** are the regions where the diaphysis and epiphyses meet. The *epiphyseal (growth) plate*, a cartilaginous structure that lies between the epiphyses and metaphyses, is present only during childhood and serves as the site of longitudinal growth. When growth ceases, the growth plate is replaced with mature bone, forming the *epiphyseal line*.
- The **periosteum** is a thin layer of connective tissue that covers and protects long bones but does not cover joint surfaces.

The knee joint is the site where the femur and tibia meet. According to the passage, OA can result in the formation of osteophytes (bony growths) on the subchondral bone of joints, which is the bone underlying the damaged articular cartilage. Because the subchondral bone of knee joints would be found at the *ends* of long bones (eg, femur), OA would likely cause additional bone **deposits on the epiphyses** of these bones.

(Choices A and C) Beneath the periosteum lies cortical (compact) bone, the hard and dense outer layer of bone that surrounds cancellous (spongy) bone. Spongy bone is the less-dense,

inner layer of bone that forms a porous, interconnected network of irregular fine bones called trabeculae. The "pores" between trabeculae are filled with red bone marrow whereas yellow marrow is found mostly in the medullary cavity. However, *nothing* in the passage suggests that OA causes deterioration of yellow marrow or loss of hemopoietic function in red marrow.

(Choice D) OA is a disease that affects joints lined with articular cartilage, which would be found at the *ends* of affected bones. Therefore, the epiphyses, *not* the diaphyses, of long bones are affected.

Educational objective:

The structure of a long bone includes the epiphyses (rounded ends covered by articular cartilage), the diaphysis (shaft containing the medullary cavity filled with yellow bone marrow), and the metaphysis (where the diaphysis and epiphyses meet).

Time spent:0 sec

Subject:
Biology

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System:
Musculoskeletal System